Forecasting Suitable Travel Periods for North Iceland
Comparison of Random Forest and LSTM Models

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Winchester, 09/03/2026

ABSTRACT

Iceland has become a major destination for nature-based tourism due to its volcanic landscapes, geothermal phenomena, glaciers, waterfalls and black sand beaches [1]. As many attractions are highly dependent on seasonal and weather conditions, travel planning is strongly influenced by climate variability.

This study aims to identify favourable travel periods between 1 March 2026 and 28 February 2027 for selected tourism activities in North Iceland. Weather conditions are forecast using two approaches — Random Forest and Long Short-Term Memory (LSTM) — trained on historical meteorological data and evaluated to determine which model provides the most reliable predictions. The objective is to estimate periods most suitable for tourism activities.

INTRODUCTION

Iceland has experienced rapid tourism growth, with visitor numbers rising from fewer than 500,000 in 2010 to over two million by 2019, and around 1.7 million annually today [1][2]. Despite covering 103,000 km², the country has a small population concentrated mainly around Reykjavík [3][4]. Located on the Mid-Atlantic Ridge, Iceland is characterised by volcanic and geothermal activity [5]. Although near the Arctic Circle, its climate is moderated by the Gulf Stream and remains highly variable due to Atlantic–Arctic air interactions [6].

This study focuses on North Iceland, particularly the Akureyri region, where severe weather may restrict access to tourist sites. Akureyri, often considered the capital of North Iceland, has about 19,000 residents and offers cultural and recreational attractions such as museums, geothermal baths, a botanical garden and outdoor activities [7][8]. The local tundra climate (≈0.7 °C annual temperature; 1261 mm precipitation) makes many activities weather dependent, including the Diamond Circle route, Northern Lights observation, whale watching, skiing and golf [7]. This study aims to identify the most suitable travel periods to Iceland between 1 March 2026 and 28 February 2027 for weather-dependent tourism activities, while activities largely unaffected by weather are excluded.

The task is formulated as a **regression-based time-series forecasting problem**, predicting future meteorological variables (temperature, precipitation, cloud cover and wind speed) using Random Forest and Long Short-Term Memory (LSTM) models. The better-performing model is then used to identify favourable travel periods for the selected activities.

PROBLEM STATEMENT

Iceland offers numerous natural attractions, many of which are strongly influenced by seasonal and weather conditions. At the same time, the country has one of the highest price levels in Europe [9], meaning that travel involves considerable cost while weather uncertainty creates risks for planned activities, such as cloud cover preventing Northern Lights observation or severe weather causing temporary road closures.

This study therefore analyses historical data to estimate the periods between 1 March 2026 and 28 February 2027 when favourable conditions for selected tourism activities are more likely to occur.

Research objective

The objective of this study is to forecast suitable travel periods for North Iceland using meteorological data and to compare the performance of Random Forest and LSTM models in time-series prediction.

DATASET

The dataset [10] was obtained from the Open-Meteo Historical Weather API, which provides reanalysis-based meteorological data derived from multiple observational sources. It contains hourly weather observations for North Iceland from 1 January 2000 to 15 February 2026, based on coordinates near Akureyri. Variables include temperature, precipitation, cloud cover, wind speed, gusts, wind direction, snowfall and WMO weather codes. Sunrise and sunset data were used to determine night-time periods. The data were downloaded in three segments and merged using the time variable.

METHODOLOGY

Activity Conditions

To estimate favourable travel periods for specific tourism activities, objective meteorological thresholds must first be defined. These thresholds determine whether an activity is feasible under given weather conditions and are later translated into model features indicating potential availability.

Northern Lights (Aurora Borealis)

The Northern Lights occur when charged particles interact with the Earth's magnetosphere [16]. Although auroral activity is commonly measured using the Kp-index (0–9), this variable is not included in this study. In Iceland, visibility mainly depends on cloud cover [17].

Auroras are typically visible from late August or early September to mid-April [18][10]. This study defines the observation period as 1 September–15 April. Visibility requires night-time conditions and cloud cover below 50% (4/8 okta) [20], excluding full moon nights due to increased natural illumination [21].

Diamond Circle and Hiking Routes

The Diamond Circle connects major attractions in North Iceland, including Dettifoss, Goðafoss, Mývatn, Ásbyrgi and Húsavík [11]. Accessibility is strongly affected by weather. Roads may close when sustained wind speeds reach 19 m/s or gusts exceed 26 m/s [12], or when snowfall and winds above 10 m/s create snowdrifts in winter [13]. Heavy rainfall above 4 mm/hour may also reduce access [14]. Route 864 to Dettifoss is typically closed between 1 October and 31 May [15].

Whale Watching

Whale watching is a major tourism activity in North Iceland, with humpback and minke whales and dolphins frequently observed [22]. Tour operation mainly depends on sea conditions rather than whale presence. Strong winds may cause cancellations; therefore wind speed is evaluated using the Beaufort scale, with force 6 (≈38 km/h) considered critical [23][24][25].

Ski Centre

Ski centre operation depends mainly on snow conditions [26]. The skiing season is assumed to extend from mid-November to mid-March. Strong winds may cause slope closures; therefore Beaufort force 6 (≈38 km/h) is used as the operational threshold.

Golf Course

Golf courses in Iceland are typically open from mid-May to late October [27]. This study defines the operational season as 15 May–31 October. Weather mainly affects playing comfort rather than accessibility, therefore weather variables are excluded from the analysis.

Summary of Criteria
Table 1. Activity Criteria Table.
Source: Author’s own compilation (see Appendix 1).

Aurora	**Whale watching**	**Ski Resort**	**Golf Course**	**Diamond Circle – wind intensity**	**Diamond Circle – snowdrift**	**Mud**	**Seasonal closure – Road 864**
<50							
sunset<sunrise							
01/09/…. T00:00 <16/04/….T00:00 +1		15/11/…. T00:00 <16/03/…. T00:00 +1	15/05/….T00:00 <01.11/….T00:00		01/10/….T00:00 01/05/….T00:00 +1	01/04/….T00:00 01/10/….T00:00	01/10/….T00:00 01/06/….T00:00 +1
0							
	225<315						
	<38	<38		<68.4	<36.0		
					0	<4	
or	or	or	or	or	and	and	or

EDA

Exploratory data analysis (EDA) examined statistical properties and seasonal patterns of the meteorological variables relevant to the predictive models. Three Excel datasets were merged and standardised, with the time variable converted to datetime and daily observations transformed into hourly records. The dataset was checked for missing values, duplicates and continuity, while distribution analysis assessed variable characteristics.

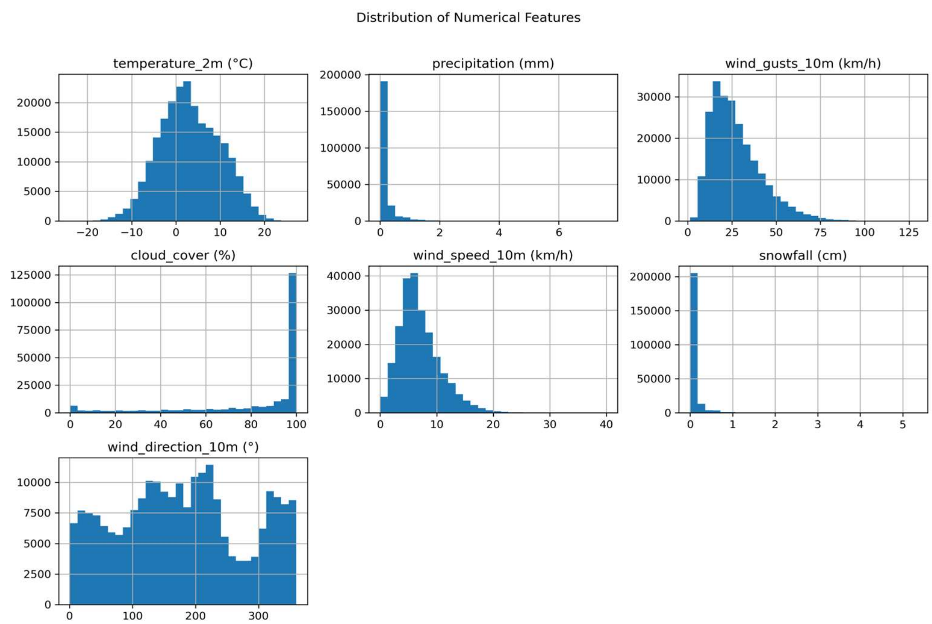

Figure 1. Distribution of numerical features.
Source: Author's own visualisation based on dataset [10]

Boxplots indicate the presence of outliers in several variables; however, these values are neither extreme nor meteorologically implausible, therefore aggressive outlier removal was not considered necessary.

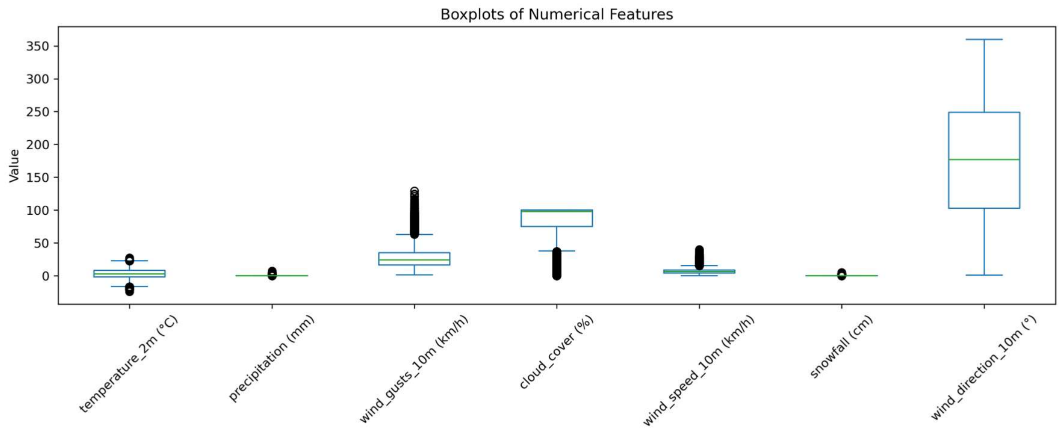

Figure 2. Boxplots of numerical features.
Source: Author's own visualisation based on dataset [10].

Correlation matrix: A strong positive relationship is observed between precipitation and snowfall, as well as between wind speed and wind gusts, while the remaining correlations are generally weak.

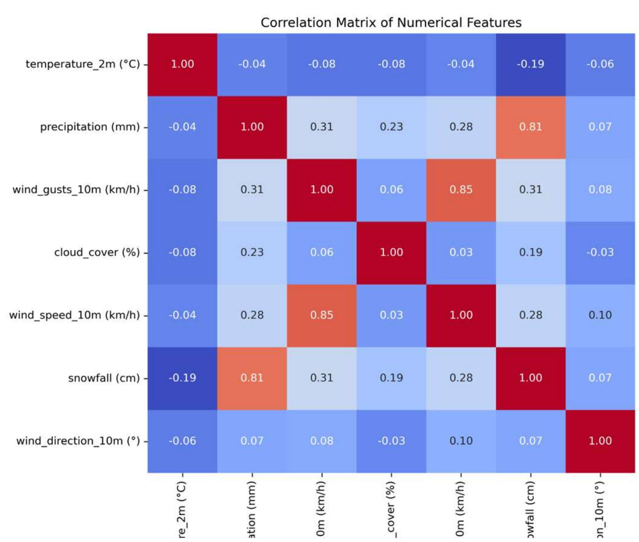

Figure 3. Correlation matrix of numerical features.
Source: Author's own visualisation based on dataset [10].

Multicollinearity can make regression estimates less reliable and harder to interpret by obscuring the individual effects of correlated variables [28]. As strong correlations were observed between precipitation and snowfall, and between wind speed and wind gusts, snowfall and wind gust variables were excluded from the analysis.

Seasonal Decomposition

Seasonal decomposition indicates that most meteorological variables — including temperature, precipitation, snowfall, wind speed and cloud cover — exhibit clear annual seasonality, while their long-term trends remain broadly stable. Snowfall is concentrated in winter months, reflecting strong seasonal dependence. Wind direction shows high variability and clustering around southerly directions, although interpretation is limited due to the circular nature of the variable (0–360°). Wind gusts display notable variability with a slight decreasing trend over time.

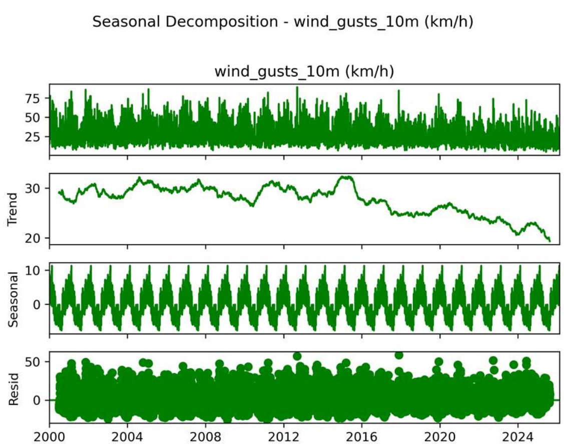

Figure 6. Seasonal decomposition of wind gusts.
Source: Author’s own visualisation based on dataset [10].

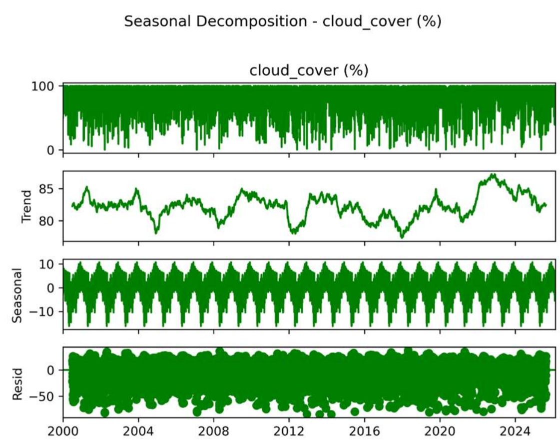

Figure 7. Seasonal decomposition of cloud cover.
Source: Author’s own visualisation based on dataset [10].

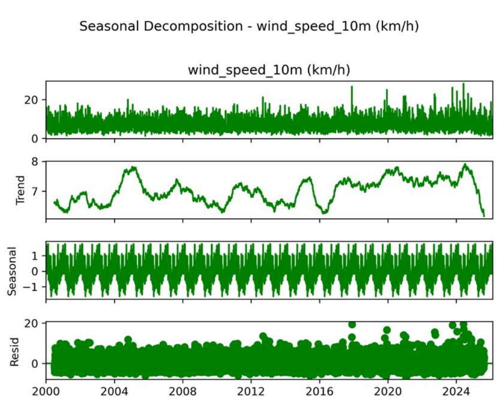

Figure 8. Seasonal decomposition of wind speed.
Source: Author’s own visualisation based on dataset [10].

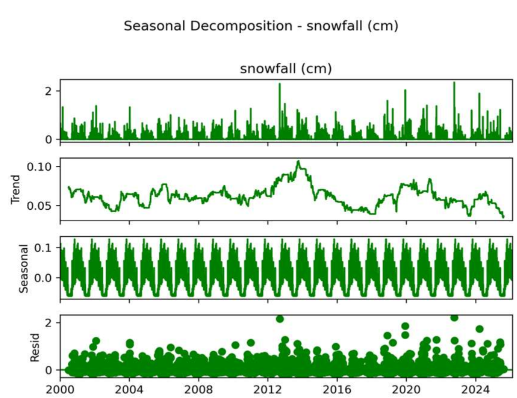

Figure 9. Seasonal decomposition of snowfall.
Source: Author’s own visualisation based on dataset [10].

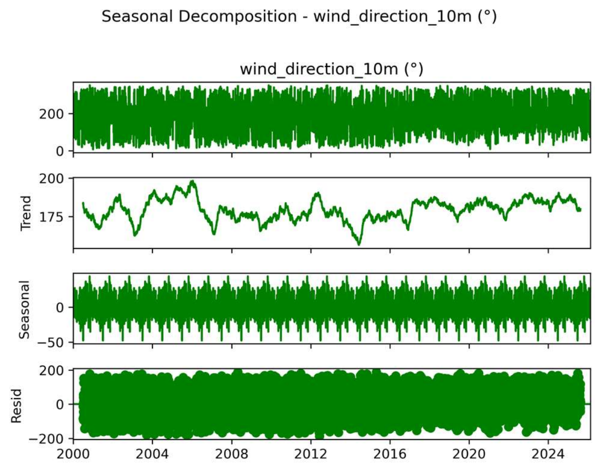

Figure 10. Seasonal decomposition of wind direction.
Source: Author’s own visualisation based on dataset [10].

Model Selection

Model selection addressed multicollinearity identified in the correlation matrix. Snowfall and wind gust variables were removed due to strong correlations with precipitation (81%) and wind speed (85%), improving model stability and interpretability.

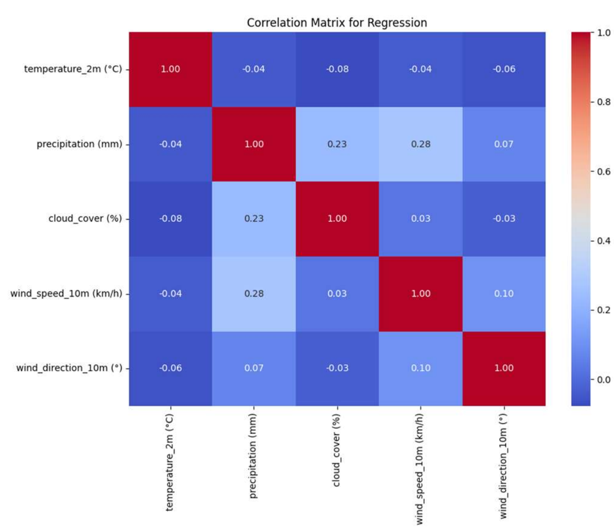

Figure 11. Correlation matrix of the modified dataset.
Source: Author’s own visualisation based on dataset [10].

Machine Learning Model Selection

Polynomial regression was applied to identify the most appropriate functional form for modelling long-term trends in the meteorological variables. Testing different polynomial degrees showed that low-order polynomials (2–3) were insufficiently flexible, while 4th- and 5th-degree polynomials provided the best fit for capturing non-linear trend variations [29]. Alternative approaches, including linear regression, logistic regression, ridge and lasso regression, SVR, ARIMA/SARIMA and GAM, were excluded due to their limitations in modelling non-linear patterns or handling the multivariate structure of the dataset.

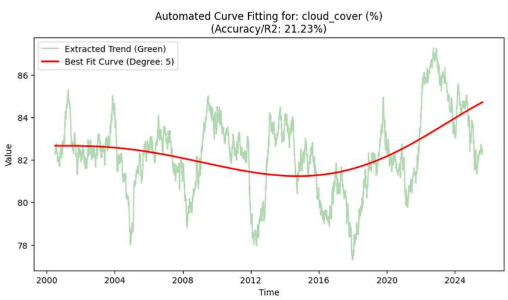

Figure 12. Best-fitting polynomial curve for cloud cover.
Source: Author’s own visualisation based on dataset [10].

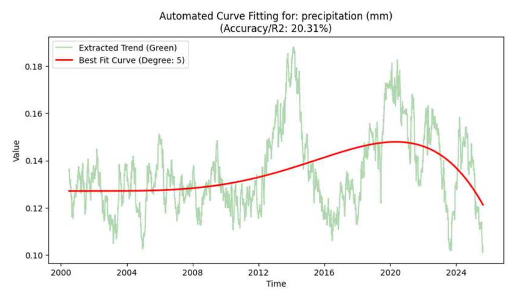

Figure 13. Best-fitting polynomial curve for precipitation.
Source: Author’s own visualisation based on dataset [10].

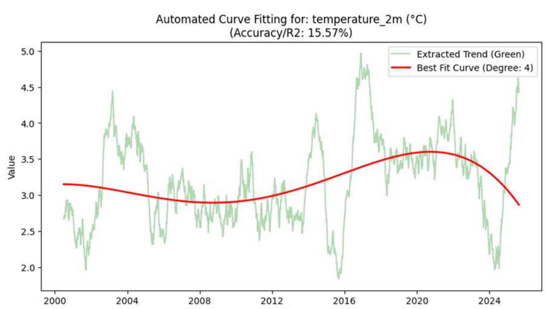

Figure 14. Best-fitting polynomial curve for temperature.
Source: Author’s own visualisation based on dataset [10].

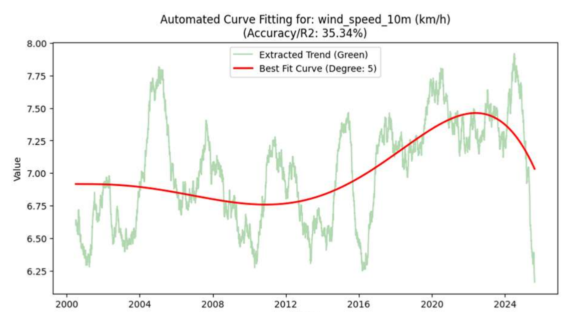

Figure 15. Best-fitting polynomial curve for wind speed.
Source: Author’s own visualisation based on dataset [10].

Selected Model

The final model selected was the **Random Forest Regressor**, a tree-based ensemble method that generates predictions by aggregating the outputs of multiple decision trees. [30]
The model is well suited for capturing non-linear relationships and interactions among variables, and demonstrated the lowest prediction error during testing. Consequently, Random Forest was chosen as the primary forecasting model.

Deep Learning Model Selection

The task involves forecasting multivariate meteorological time series, requiring a model capable of capturing temporal dependencies and non-linear relationships within large sequential datasets. Conventional feedforward neural networks (ANN/MLP), convolutional neural networks (CNN), autoencoders and generative adversarial networks (GAN) were excluded as they are not designed for temporal forecasting tasks. Classical recurrent neural networks (RNN) were also rejected due to instability when modelling long sequences, while transformer architectures were considered unnecessarily complex and computationally demanding for the scope of this study.

Selected Model

Long Short-Term Memory (LSTM) networks were selected as the deep learning model, as their specialised memory cells allow them to capture long-term temporal dependencies, making them well suited for forecasting multivariate meteorological time series. [31]

Prevention of Data Leakage

To avoid data leakage, the dataset was split strictly in chronological order. The training set covered **2000–2020**, the validation set **2021–2023**, and the test set **2024–2026**, ensuring that future observations were not used during model training. This approach guarantees that model evaluation reflects a realistic forecasting scenario based only on past information.

Random Forest Model Development

The time variable was set as the index and the dataset was ordered chronologically for time-series modelling. Lag features (1, 3 and 24 hours) and temporal variables (hour, day of year, month, and a day–night indicator based on sunrise and sunset) were created to capture temporal patterns. The data were split chronologically into training (2000–2020), validation (2021–2023) and test (2024–15 February 2026) sets to avoid future information leakage.

Separate Random Forest models were trained for each target variable using 200 trees (random_state = 42, n_jobs = -1). Model performance was evaluated using MAE and RMSE, and recursive forecasting was used to generate hourly predictions for March 2026–February 2027, with physical constraints applied (e.g. precipitation ≥ 0, cloud cover 0–100%).

LSTM Model Development

Wind direction was converted from degrees to radians and represented using sine and cosine transformations (wind_sin, wind_cos) to account for its cyclic nature. The dataset was split chronologically into training (2000–2020), validation (2021–2023) and test (2024–2026) sets, containing 184,103, 26,280 and 18,648 observations respectively. Input variables were normalised using MinMaxScaler fitted on the training data, while target variables were scaled separately.

The data were transformed into time-series sequences using a 24-hour input window with eight input features and six target variables. The network consisted of an LSTM layer with 64 units, a Dropout layer (0.2) to reduce overfitting, and a Dense output layer predicting six targets (19,078 trainable parameters). The model was trained using the Adam optimiser and MSE loss, with MAE and RMSE used for evaluation.

Early stopping monitored validation loss and halted training if no improvement occurred for five epochs, restoring the best weights. Training was performed for up to 50 epochs with a batch size of 64.

RESULTS

Random Forest Results

The results in Table 2 indicate that the RMSE values remain relatively stable across the validation and test datasets. However, the training error is substantially lower (MAE ≈ 0.60; RMSE ≈ 1.30) compared with the validation (MAE ≈ 2.02; RMSE ≈ 3.10) and test sets (MAE ≈ 1.99; RMSE ≈ 3.06). This gap clearly indicates overfitting, suggesting that the model memorised the training data rather than learning the underlying patterns, which limits its generalisation ability for long-term forecasting.

Table 2. Random Forest overfitting results.
Source: Author's own visualisation based on dataset [10].

Variable	MAE Train	Train-Val diff	MAE Validation	Val-Test diff	MAE Test
temperature_2m (°C)	0.114443894	0.344108085	0.458551979	0.003883859	0.462435838
precipitation (mm)	0.018151983	0.07916086	0.097312842	0.001887608	0.09920045
cloud_cover (%)	2.354642384	9.27157375	11.62621613	0.474611569	12.1008277
wind_speed_10m (km/h)	0.25126448	0.788634816	1.039899296	0.067717687	1.107616983
wind_direction_10m (°)	6.245518742	25.63140287	31.87692161	-1.602320316	30.2746013

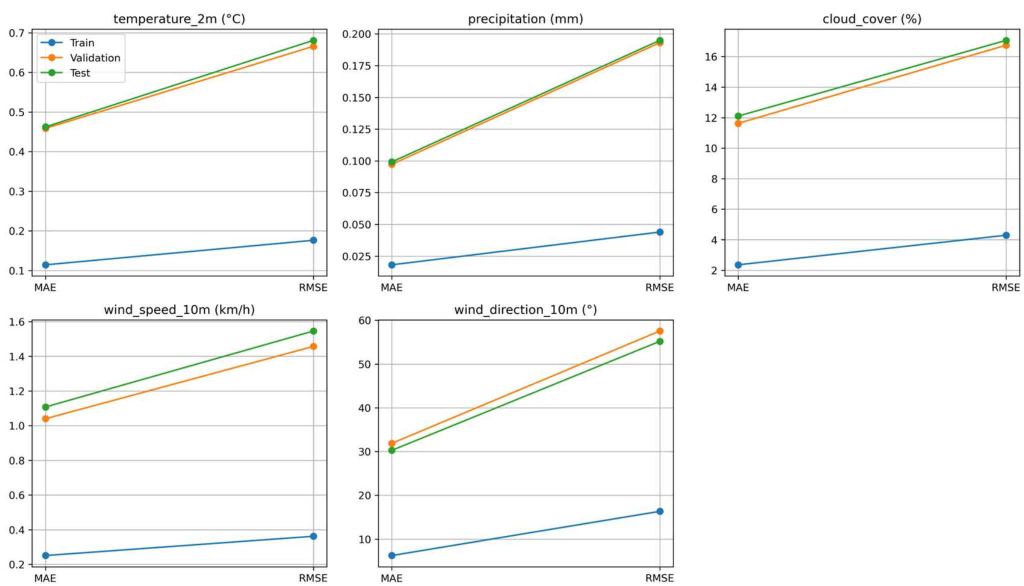

Figure 16. Overfitting observed in the Random Forest model.
Source: Author's own visualisation based on dataset [10].

Wind direction posed an additional challenge because the circular nature of the variable causes artificial jumps (e.g. 359° to 1°). To address this, wind direction was transformed into sine and cosine components, allowing the model to represent the circular relationship correctly.

Model Adjustments

Several modifications improved model stability. Short-term lags (1–3 hours) were removed and replaced with 24-hour and one-year lags. Wind direction was transformed into sine–cosine components to represent circularity. Recursive forecasting used previous-year anchors with calendar-based stepping. Random Forest parameters were constrained to reduce overfitting and improve generalisation.

2. Random Forest Results

The RMSE values show no abnormal deviations; therefore, MAE was used as the primary evaluation metric. The difference between the training MAE (2.00) and validation MAE (2.39) is moderate (~0.4), indicating that the model no longer memorises the training data and that overfitting has been substantially reduced. Model stability is further confirmed by the very small differences between validation and test errors (e.g. temperature: 0.25; precipitation: −0.003), demonstrating consistent performance on unseen data.

The sine–cosine transformation successfully resolved the circular discontinuity between 0° and 360° wind directions, enabling stable and physically meaningful predictions. This is supported by the extremely small validation–test error difference (<0.02) for wind direction. Overall, the results indicate stable model behaviour with no significant overfitting, although the remaining error levels suggest that perfect prediction of complex meteorological variables is not achievable.

Table 3. Random Forest results.

Variable	Train MAE	Train-Validation diff	Validation MAE	Validation-Test diff	Test MAE
temperature_2m (°C)	2.002557095	0.394796576	2.397353671	0.250526513	2.64788
precipitation (mm)	0.167945295	0.111684755	0.27963005	-0.003066132	0.276564
cloud_cover (%)	19.85711816	4.73477258	24.59189074	0.308929141	24.90082
wind_speed_10m (km/h)	2.367399356	0.619210766	2.986610121	0.375291545	3.361902
wind_sin	0.486473366	0.050200678	0.536674044	-0.020440237	0.516234
wind_cos	0.514692351	0.063464404	0.578156755	0.002156632	0.580313

Source: Author’s own visualisation based on dataset [10].

LSTM Results

The LSTM model achieved strong performance, estimating the target variables with approximately 4–5% average error on the validation dataset. In the final training epoch, the training loss was 0.0055 and the validation loss was 0.0086, with a validation MAE of 0.044 and RMSE of 0.0927.

Across individual variables, the model remained stable on the train, validation and test datasets. Temperature prediction errors were 0.43 °C (train), 0.57 °C (validation) and 0.56 °C (test), while precipitation errors were 0.06 mm, 0.08 mm and 0.08 mm respectively. Cloud cover errors were 6.42% (train), 8.23% (validation) and 8.36% (test), and wind speed errors were 0.71 km/h, 0.99 km/h and 1.06 km/h.

The cyclic wind-direction variables also showed low errors (wind_sin: 0.09–0.14; wind_cos: 0.09–0.13). Overall, the small differences between train, validation and test results indicate good generalisation, and the model shows no evidence of overfitting.

Table 4. LSTM results.
Source: Author's own visualisation based on dataset [10].

feature	train_mae	train_validation_diff	validation_mae	validation_test_diff	test_mae
temperature_2m (°C)	0.432414558	0.134590204	0.567004762	0.005319194	0.561685568
precipitation (mm)	0.061559531	0.017120544	0.078680074	0.000378053	0.078302021
cloud_cover (%)	6.42363887	1.806441629	8.230080499	0.129753703	8.359834202
wind_speed_10m (km/h)	0.708806766	0.276363048	0.985169814	0.07663045	1.061800264
wind_sin	0.092371451	0.04508442	0.137455871	6.07559E-05	0.137395115
wind_cos	0.088305198	0.03790342	0.126208618	0.000750824	0.126959442

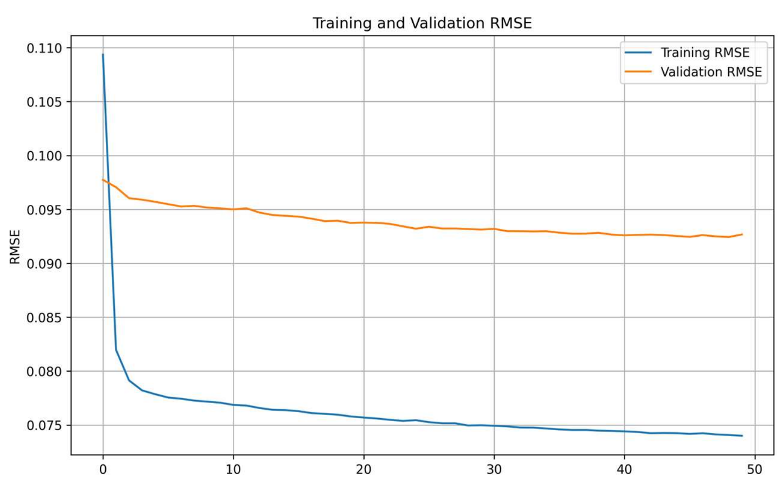

Figure 17. LSTM training and validation RMSE during model training.
Source: Author's own visualisation based on dataset [10].

Model Comparison

The results show that the LSTM model substantially outperforms the Random Forest regression model across the evaluated variables. For example, temperature prediction error decreases from approximately 2.39 °C (Random Forest) to about 0.57 °C with LSTM, while precipitation and wind speed errors also remain low (around 0.08 mm and ~1 km/h respectively). The small differences between training, validation and test results indicate good generalisation without significant overfitting, and the model also performs very well on the previously unseen test data.

This improved performance is mainly due to the LSTM architecture, which is specifically designed for time-series data and can capture long-term temporal dependencies through its internal memory mechanism. In contrast, Random Forest treats observations largely as independent samples and incorporates temporal information only through predefined lag features. Consequently, the LSTM model better captures the temporal dynamics of meteorological data and was therefore selected as the final forecasting model.

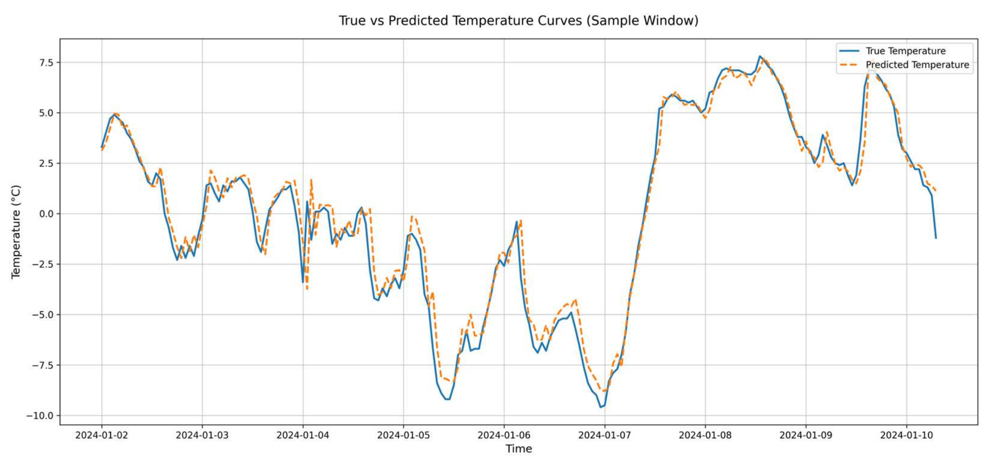

Figure 18. LSTM true vs predicted curves on the test dataset.
Source: Author’s own visualisation based on dataset [10].

Forecast-Based Identification of Suitable Travel Periods

Forecasted weather variables were used to identify periods when selected tourism activities could realistically take place. Activity-specific environmental and seasonal criteria reflecting operational constraints were applied to hourly forecasts.

Whale watching required wind speeds below 38 km/h and favourable wind directions. Northern Lights observation required night-time conditions, cloud cover below 50% and the aurora season. The Diamond Circle required low precipitation and moderate wind, while skiing was limited to winter.

Suitable hours were aggregated to daily values and grouped into continuous tourism windows.

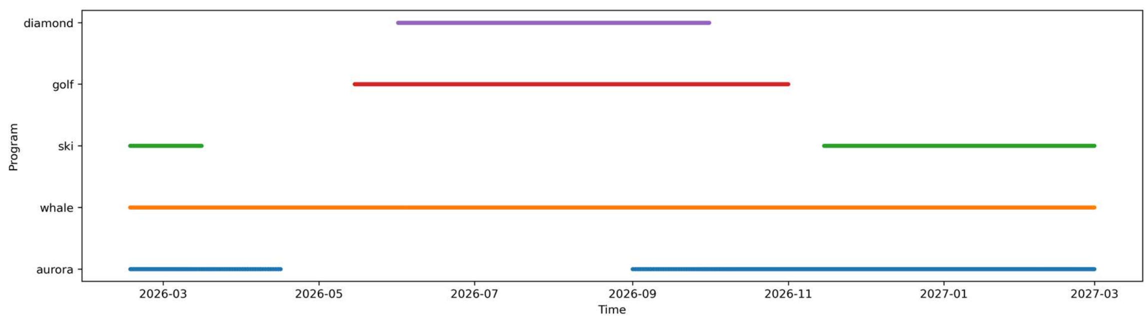

Figure 18. Program availability based on forecasted weather conditions.
Source: Author's own visualisation based on dataset [10].

Forecast Results

The forecast indicates that visitors interested in the Diamond Circle, whale watching and golf can travel between **1 June and 30 September 2026**, when at least four suitable days per week are expected for these activities.

If Northern Lights observation is also desired, the most favourable period is **16 February – 16 March 2026**. For visitors wishing to combine **skiing, whale watching and aurora observation**, the most suitable travel windows are **16 February – 15 March 2026** or **15 November 2026 – 28 February 2027**, when winter conditions support both skiing and a high probability of aurora visibility.

DISCUSSION

The results indicate that deep learning models capture temporal patterns in meteorological time series more effectively than traditional machine learning approaches. The Random Forest model showed signs of limited generalisation, whereas the LSTM model demonstrated consistent performance across the training, validation and test datasets, indicating reliable predictive capability. This improvement is likely related to the ability of LSTM networks to retain temporal dependencies through their internal memory structure.

The forecast-based activity analysis also illustrates how meteorological predictions can be translated into practical insights for tourism planning.

CONCLUSION

This study aimed to evaluate the ability of machine learning and deep learning models to forecast meteorological variables and support tourism planning in Iceland. The results show that while the Random Forest model produced reasonable predictions, its accuracy remained limited, as the model mainly captured broader trends in the data rather than the underlying temporal dependencies. In contrast, the LSTM model achieved lower prediction errors and demonstrated stable performance across the training, validation and test datasets, indicating that deep learning architectures are more effective for modelling complex temporal patterns in meteorological time series.

Using the selected model, it was possible to estimate periods when tourism activities in Iceland are likely to be feasible based on predicted weather conditions. The forecast suggests that travel between **1 June and 30 September 2026** is likely to provide favourable conditions for selected activities. If Northern Lights observation is also desired, a favourable period is **16 February – 16 March 2026**. Another suitable travel window is **15 November 2026 – 28 February 2027**, when there is also a high probability that the selected activities can be undertaken successfully.

Limitations

The model estimates the meteorological suitability of tourism activities rather than their actual occurrence. As the forecasts are based on weather conditions in Akureyri and North Iceland, the results cannot be generalised to the entire country. Tourism activities may also depend on operational decisions, safety regulations and local conditions.

Several simplifications were applied: wind gusts were combined with wind speed, snowfall with precipitation, and Northern Lights visibility estimated without KP-index data. Seasonal restrictions were also applied to routes such as the Diamond Circle.

Finally, weather forecasts involve inherent uncertainty; therefore the identified tourism windows represent probabilistic estimates rather than exact predictions.

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Appendix 1 – Activity Criteria Table

Aurora	Whale watching	Ski Resort	Golf Course	Diamond Circle – wind intensity	Diamond Circle – snowdrift	Mud	Seasonal closure – Road 864
<50							
sunset<sunrise							
01/09/…. T00:00 <16/04/….T00:00 +1		15/11/…. T00:00 <16/03/…. T00:00 +1	15/05/….T00:00 <01.11/….T00:00		01/10/….T00:00 01/05/….T00:00 +1	01/04/….T00:00 01/10/….T00:00	01/10/….T00:00 01/06/….T00:00 +1
0							
	225<315						
	<38	<38		<68.4	<36.0		
					0	<4	
or	or	or	or	or	and	and	or